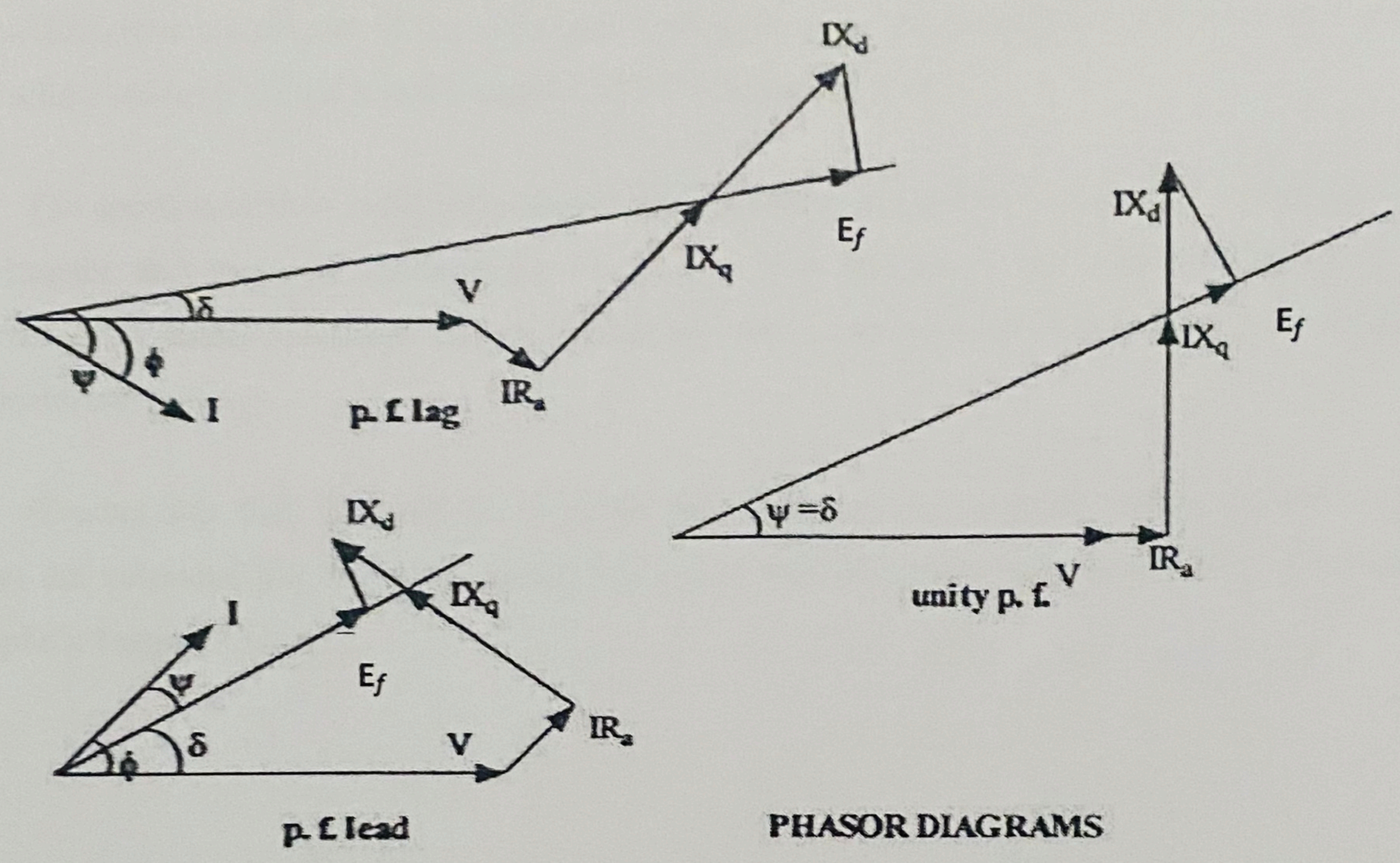
minimum. Constant applied voltage minus the minimum impedance voltage drop (armature current being minimum) in the leads and 3-phase variac gives maximum armature-terminal voltage. The ratio of maximum armature terminal voltage per phase to minimum armature current per phase gives Z_{sd}. After one quarter of slip cycle, the peak of armature mmf is in line with q-axis and the reluctance offered by the magnetic circuit is maximum. The armature current, required for the establishment of constant air-gap flux, will be maximum and the armature terminal voltage will be minimum. The ratio of minimum armature terminal voltage per phase to maximum armature current per phase gives Zsq.



PHASOR DIAGRAMS

When the armature mmf is in line with field poles, the armature flux linkage with field winding is maximum and rate of change of this flux linkage is zero, so that induced voltage across the field winding is zero. On the other hand, when armature mmf is in line with q-axis, the flux linkage with field winding is minimum and rate of change of this flux linkage is maximum, so that induced voltage across the field winding is maximum.

PROCEDURE

SLIP TEST

Make the connections as shown in figure.

Precautions

i) Keep the autotransformer at minimum voltage position

ii) Keep DPST, TPST and SPST switches open

iii) Keep dc motor field rheostat at minimum resistance position